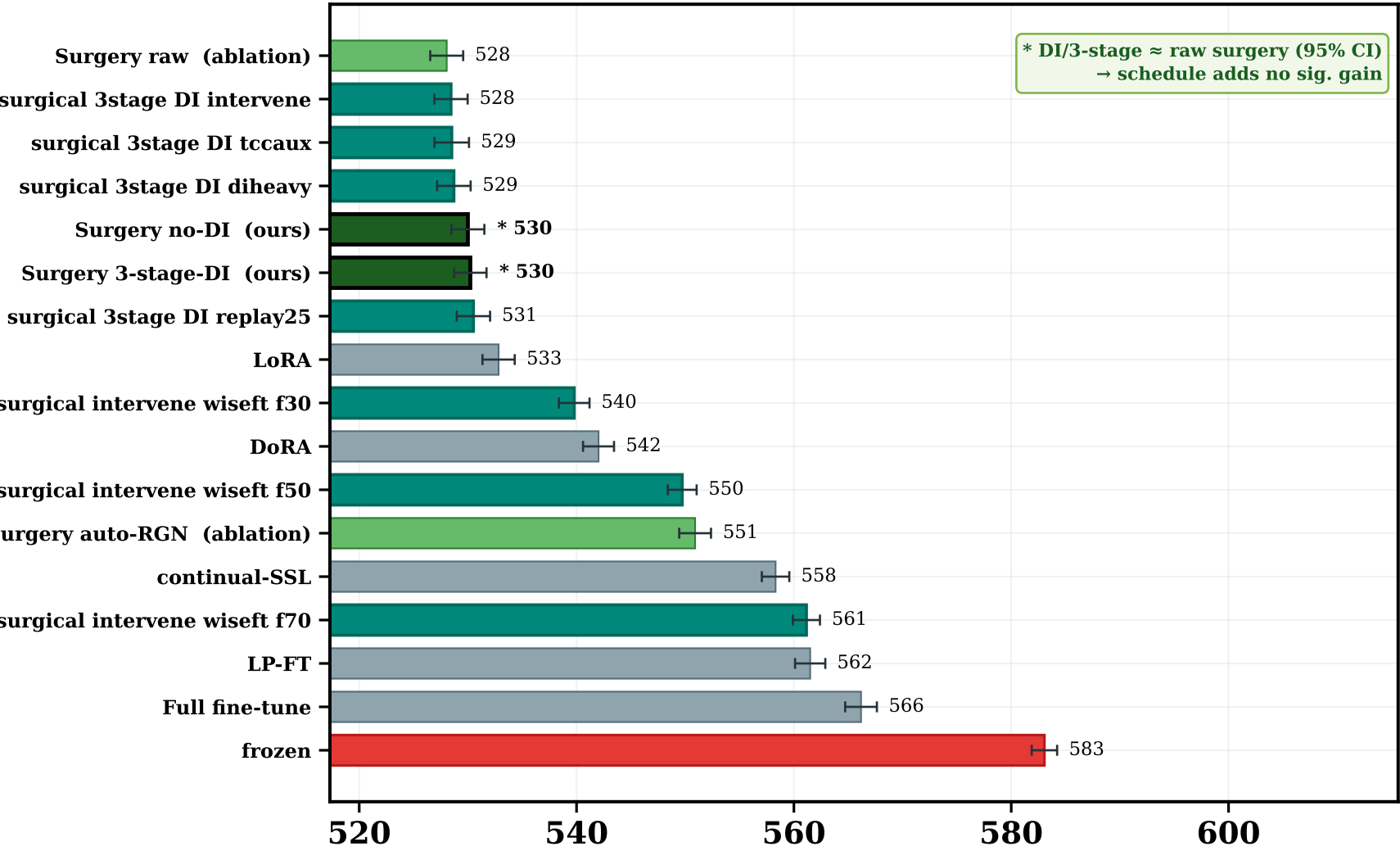
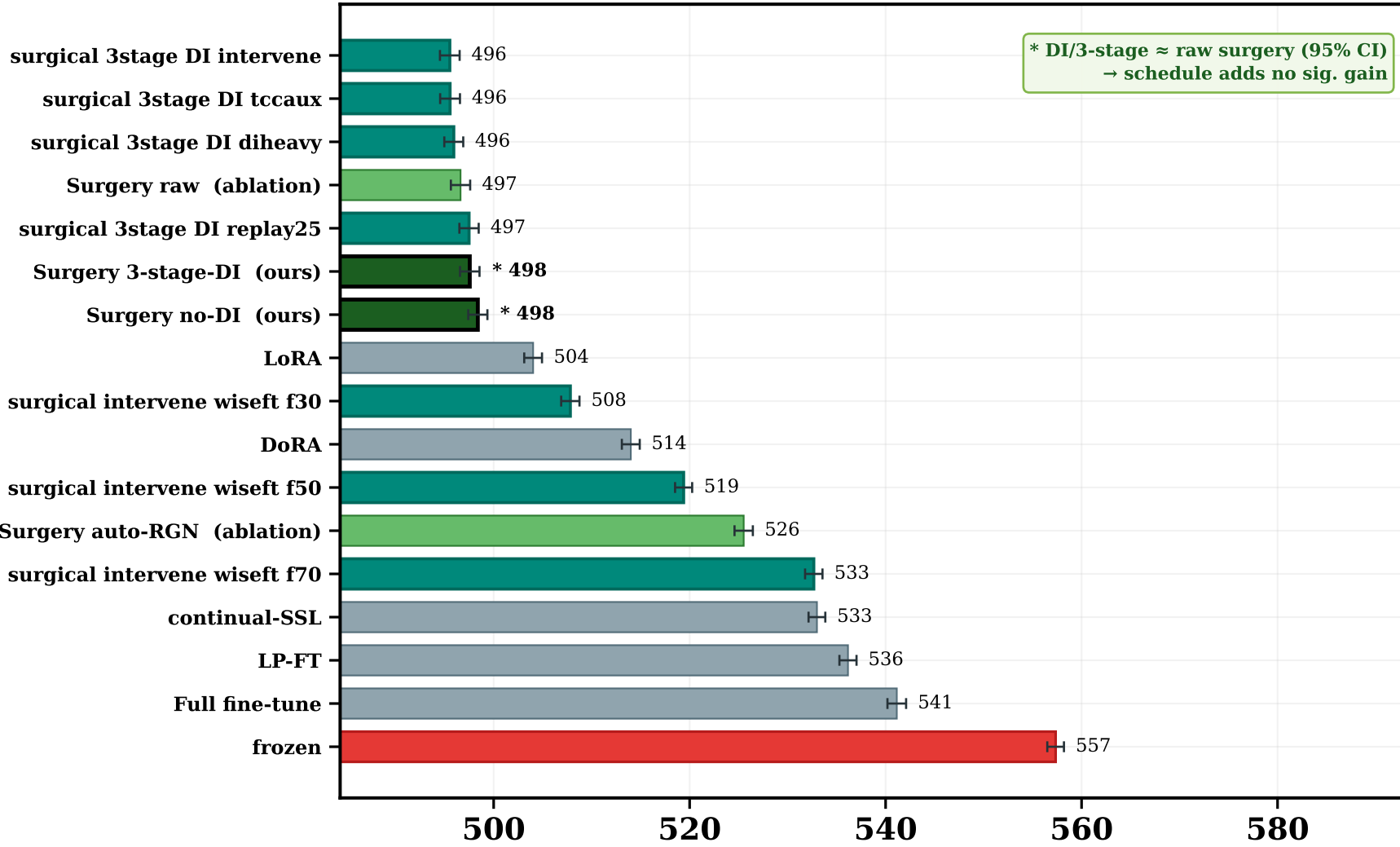


Factor surgery dominates predictive-quality metrics — full baseline set, no per-panel hiding (* = ours · green band = surgery family · 95% BCa CI)

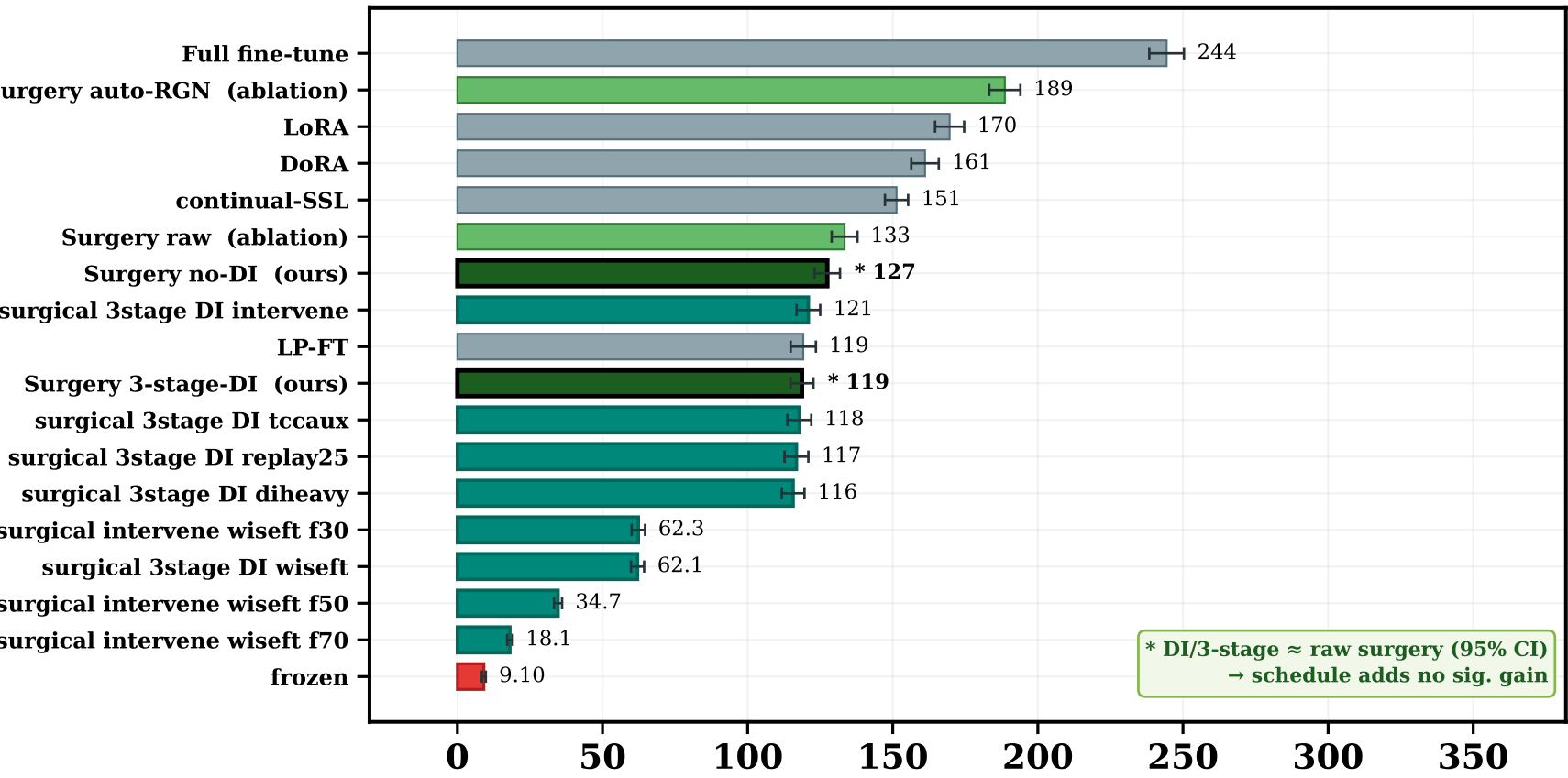
HERO · Causal L1 (sensitivity to causal perturbation) (↓ lower better) (×10⁻³)



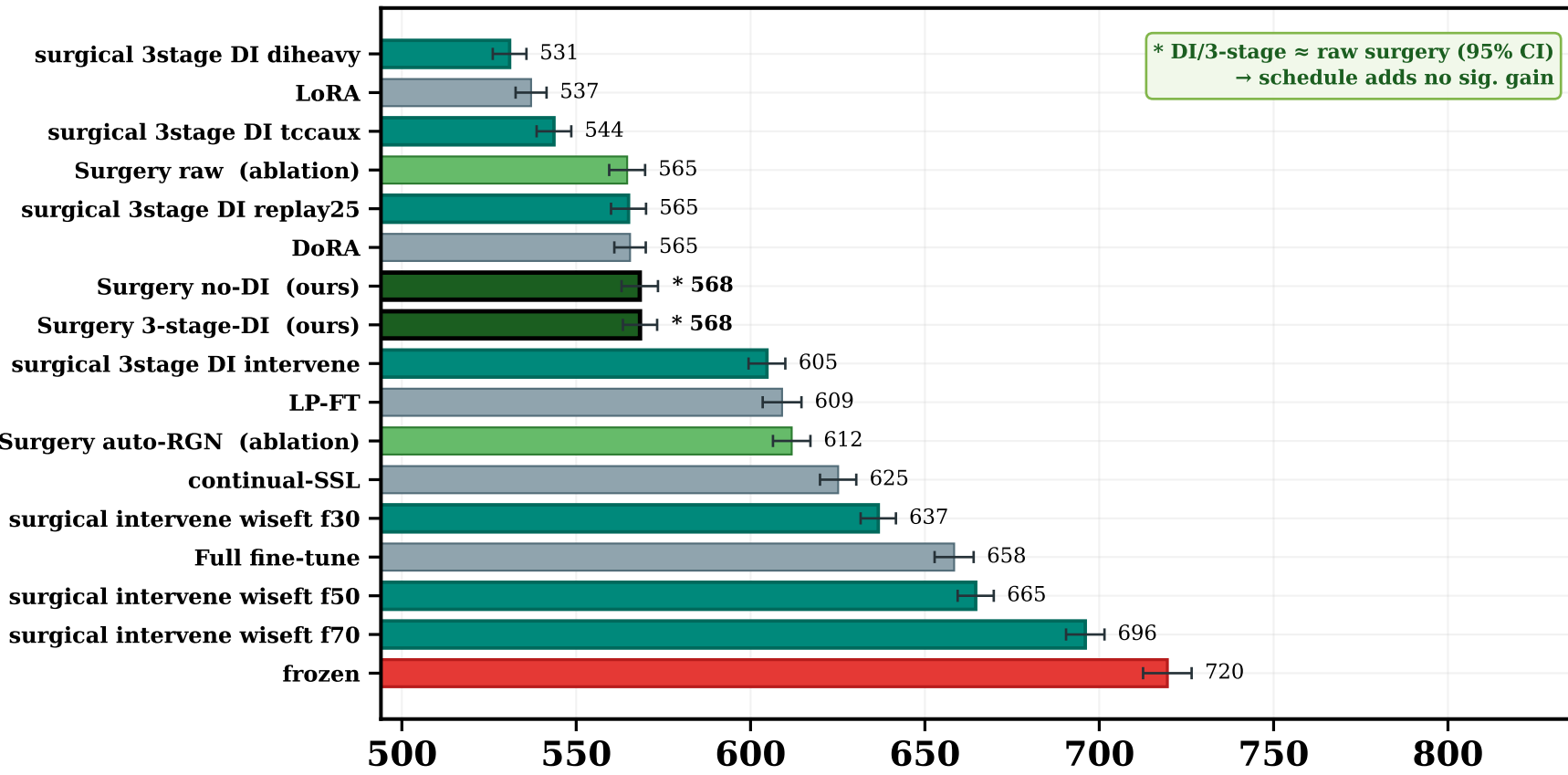
HERO · Future-frame MSE (world-model prediction) (↓ lower better) (×10⁻³)



Motion-cosine (motion coherence) (↑ higher better) (×10⁻³)



Mask-ratio slope (masking robustness) (↓ lower better) (×10⁻⁴)



Surgery — ours (3-stage-DI, no-DI) **Fine-tune / PEFT / continual-SSL baselines** **Frozen V-JEPA (the baseline to beat)** **Surgery — iter18 improvement arms**
Surgery — ablations (raw, auto-RGN)

HONEST READ (full baseline set — nothing hidden): the FACTOR-SURGERY FAMILY occupies the top of every predictive-quality metric — causal sensitivity and future-frame MSE — with 95% CIs clear of full fine-tune, LoRA/DoRA and the frozen encoder.
The DI / 3-stage schedule (ours) MATCHES raw surgery within 95% CI: an honest ablation, not a hidden weakness — the gain comes from surgery itself. Surgery trades a small temporal-ordering / TCC margin (see full 14-panel scorecard) for this predictive lead; we report that trade-off rather than dropping those panels.